Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Lesson 1: Model Selection Techniques

- Introduction to Model Selection
  Reading • 5 min
- Bayesian Model Selection
  Reading • 30 min
- Bayesian Model Selection Readings
  Reading • 1h

Lesson 2: Cross Validation

- Introduction to Cross-Validation
  Reading • 25 min
- **K-Fold Cross-Validation**
  Reading • 2 min
- Leave-One-Out Cross-Validation (LOOCV)
  Reading • 10 min

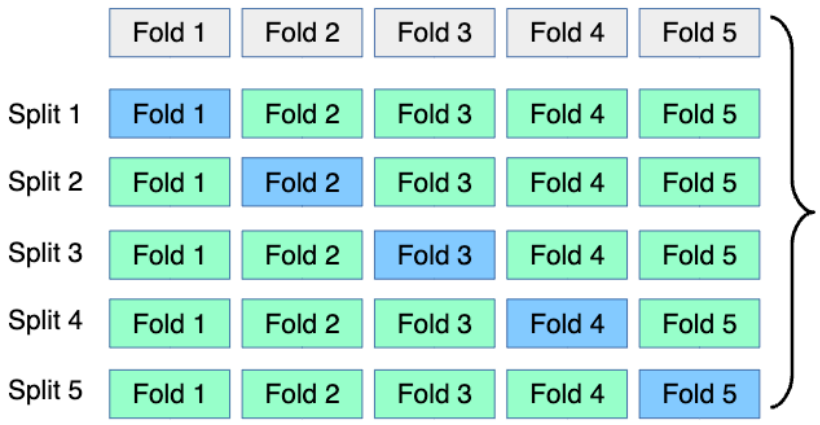
Lesson 3: Introduction to Optimization Techniques

- Introduction to Optimization Techniques
  Reading • 13 min
- Optimization Using Gradient Descent
  Reading • 40 min
- Gradient Descent with Momentum
  Reading • 45 min
- Stochastic Gradient Descent (SGD)
  Reading • 40 min
- Constrained Optimization
  Reading • 16 min
- Lagrange Multipliers
  Reading • 8 min
- Convex Optimization
  Reading • 8 min
- Linear Programming
  Reading • 15 min
- Quadratic Programming
  Reading • 10 min
- Check Your Knowledge: Optimization Techniques
  Graded Assignment • 15 min

K-Fold Cross-Validation

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K-Fold Cross-Validation is a method where the dataset is divided into K equal parts or folds. The model is trained K times, each time using a different fold as the validation set and the remaining K-1 folds as the training set. After all K iterations, the results are averaged to provide a final performance estimate. This approach efficiently uses the data by ensuring every data point is used for both training and validation, and it reduces the variance of the performance estimate.



5 fold cross validation

Steps:

1. **Divide the Data:** Split the entire dataset into K equal (or nearly equal) folds.
2. **Train the Model:** For each fold, train the model using the K-1 folds as the training data.
3. **Validate the Model:** Use the remaining fold as the validation data to test the model's performance.
4. **Repeat:** Repeat the process K times, each time with a different fold as the validation set.
5. **Average the Results:** Compute the average performance across all K iterations to get a more reliable estimate of the model's generalization error.

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